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Mass surveillance in the age of AI-the Indian dilemma of security versus privacy

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Abstract

India's rapid adoption of AI-powered surveillance technologies, from facial recognition systems to predictive policing algorithms, has created a critical tension between national security imperatives and fundamental privacy rights. This paper examines the legal, ethical, and societal implications of mass surveillance in India, contrasting its approach with global models while highlighting alarming gaps in accountability. Unlike Western democracies with robust oversight mechanisms or authoritarian states with centralized control, India operates in a regulatory vacuum, deploying advanced surveillance tools without adequate safeguards. Case studies reveal systemic issues: Misidentification by facial recognition systems leading to wrongful arrests, predictive algorithms reinforcing caste and religious biases, and the normalization of surveillance in public spaces altering citizen behavior.

The analysis demonstrates how India's legal framework, anchored in the Puttaswamy judgment's privacy protections but undermined by vague laws like IT Act Section 69, fails to meet international standards. Comparative assessments with the EU's GDPR, U.S. Fourth Amendment protections, and China's Social Credit System expose India's dangerous hybrid trajectory: authoritarian-scale surveillance capabilities with democratic-era accountability deficits.

Keywords: AI-Powered, Misidentification, predictive policing algorithms, regulatory vacuum

Introduction

The rapid advancement of artificial intelligence (AI) has ushered in a new era of mass surveillance capabilities, presenting nations worldwide with complex challenges at the intersection of security, privacy, and civil liberties. Nowhere is this tension more pronounced than in India, where the government's increasing reliance on AI-powered surveillance tools has sparked intense debate about the appropriate boundaries between state security imperatives and fundamental rights. As the world's largest democracy with over 1.4 billion citizens, India's approach to balancing these competing interests carries significant implications not just domestically but for global discourse on digital governance. The proliferation of facial recognition systems, predictive policing algorithms, and large-scale biometric databases has transformed India's surveillance landscape in recent years, often outpacing the development of corresponding legal safeguards and oversight mechanisms. This introduction examines the emergence of AI-driven surveillance in India, contextualizes it within global trends, analyzes the key technologies being deployed, and outlines the pressing legal and ethical concerns that demand urgent scholarly and policy attention.

India's journey into AI-enabled surveillance gained substantial momentum following the 2008 Mumbai terrorist attacks, which exposed critical gaps in the nation's security infrastructure. In response, successive governments have invested heavily in surveillance technologies, justifying them as necessary tools for national security, crime prevention, and efficient governance. The Aadhaar biometric identity system, launched in 2009, marked a watershed moment, creating the world's largest biometric database with over 1.3 billion enrolled Indians. While initially framed as a mechanism to streamline welfare delivery and reduce fraud, Aadhaar's scope has expanded dramatically, becoming the foundational infrastructure for an increasingly sophisticated surveillance ecosystem. This expansion has occurred alongside the deployment of facial recognition systems by police forces across major cities, the establishment of integrated command and control centers in smart cities, and the development of predictive policing algorithms all with minimal public consultation or legislative debate.

The constitutional dimensions of this surveillance surge came into sharp focus with the landmark 2017 Supreme Court judgment in Justice KS Puttaswamy v. Union of India,

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which unequivocally recognized the right to privacy as a fundamental right under Article 21 of the Indian Constitution. The ruling established a three-pronged test for state interference with privacy, legality, necessity and proportionality, balanced approach that has profound implications for AI surveillance programs. However, in practice, many surveillance initiatives appear to fall short of these standards, operating under vague legal authorities or executive orders rather than specific parliamentary legislation. The subsequent draft Personal Data Protection Bill (PDPB) of 2019^[4], intended to create a comprehensive privacy framework, has languished in legislative limbo, leaving India without modern legal safeguards comparable to the European Union's General Data Protection Regulation (GDPR) or even the more limited protections found in U.S. law.

Globally, the deployment of AI surveillance presents a spectrum of approaches that provide instructive comparisons for India's experience. At one extreme lies China's social credit system, which combines facial recognition, big data analytics, and AI scoring mechanisms to create an unprecedented architecture of social control. At the other end are democratic nations like Germany and Canada, which have implemented strict limitations on facial recognition in public spaces amid concerns about rights violations. The United States occupies a middle ground, with significant surveillance capabilities but somewhat stronger (though imperfect) judicial and legislative checks. India's trajectory appears concerningly ambiguous, with surveillance capabilities rivaling China's in scale but without China's overt authoritarianism, while simultaneously lacking the institutional safeguards characteristic of Western democracies. This ambiguous position creates unique risks the potential for "function creep" where technologies introduced for limited purposes expand uncontrollably, the danger of normalization of surveillance, and the particular threats to marginalized communities who may face disproportionate monitoring. The technological infrastructure underpinning India's AI surveillance state is both sophisticated and disturbingly opaque. Facial recognition systems (FRS) have been deployed by at least 18 Indian police forces, with the Delhi Police alone having conducted over 66,000 searches since 2018, often targeting protesters and political demonstrators. The National Crime Records Bureau's (NCRB) Automated Facial Recognition System (AFRS), intended to create a national database of criminal faces, raises profound questions about error rates (particularly for women and darker-skinned individuals), consent, and mission creep. Predictive policing tools like Punjab's Police Artificial Intelligence System (PAIS) claim to forecast crime hotspots but risk reinforcing existing biases in law enforcement practices. Perhaps most insidiously, projects like the Safe City Initiative in urban centers and the National Intelligence Grid (NATGRID) create interconnected surveillance networks that aggregate data from disparate sources traffic cameras, social media monitoring, financial transactions, and travel records with minimal transparency about data handling or access protocols.

The ethical and legal concerns stemming from this surveillance apparatus are manifold and profound. At a constitutional level, the tension between privacy rights (Article 21) and reasonable restrictions in the interest of security (Article 19(2)) remains unresolved in practical

application. The lack of a comprehensive data protection law means there are no clear rules governing data retention periods, purpose limitation, or individual redress mechanisms. Institutional oversight is woefully inadequate unlike the U.S. Foreign Intelligence Surveillance Court (FISC) or the U.K.'s Investigatory Powers Tribunal, India lacks specialized judicial bodies to review surveillance orders. The potential for abuse is exacerbated by India's political climate, where surveillance tools have allegedly been weaponized against journalists, activists, and opposition figures under the guise of national security. Furthermore, the intersection of surveillance technologies with India's complex social fabric raises disturbing possibilities of caste, religious, or community-based profiling, as algorithms trained on biased data may perpetuate and amplify historical discrimination.

This research paper seeks to systematically analyze these challenges through an Indian constitutional and legal lens. The analysis proceeds by first examining the existing legal framework governing surveillance in India, identifying critical gaps and inconsistencies. It then provides a detailed mapping of AI surveillance technologies currently deployed across Indian jurisdictions, assessing their technical capabilities and documented use cases. The paper critically evaluates several high-profile instances of surveillance application, from Jammu and Kashmir's internet shutdowns to the controversial Aarogya Setu contact tracing app during the COVID-19 pandemic. A comparative analysis situates India's approach relative to international standards and best practices, drawing lessons from both democratic and authoritarian contexts. The final sections propose concrete policy recommendations aimed at achieving a more balanced equilibrium between legitimate security needs and fundamental rights protections in India's emerging surveillance ecosystem.

As India stands at a crossroads between becoming a surveillance-obsessed state and developing a rights-respecting model of digital governance, this examination could not be more timely. The choices made today about regulating AI surveillance will have generational consequences for the quality of Indian democracy, the protection of minority rights, and the very meaning of privacy in the digital age. By interrogating these issues through legal, technological, and ethical frameworks, this research aims to contribute to informed public debate and evidence-based policymaking in one of the most critical domains of 21st-century governance.

The Legal Tightrope: India's surveillance laws in the AI Era

In the softly lit police control room, an officer diligently scans countless faces flashing across a screen, trying to match them with a criminal database. Nearby, a young lawyer is deep in thought, sifting through India's outdated privacy laws and contemplating how to challenge these practices in court. This scene highlights a significant contradiction in India: while our surveillance technology has advanced rapidly, the laws designed to regulate it seem stuck in a bygone era.

At the heart of this legal confusion is the Information Technology Act of 2000, a law created when most Indians were just beginning to explore the internet. Section 69 of this act permits surveillance for national security, but its broad language means it could apply to anything from

tracking terrorists to monitoring student protests. A retired high court judge, enjoying a cup of chai in his home in Delhi, comments, "It's like using a sledgehammer to crack a walnut. The law doesn't differentiate between a real threat and ordinary dissent."

Then came the landmark Puttaswamy judgment in 2017^[2], where nine Supreme Court judges recognized privacy as a fundamental right. This ruling should have transformed the landscape. In theory, any surveillance must now meet three criteria: it must be legally authorized, necessary for a legitimate purpose, and proportionate in its approach. However, in reality, "Police stations are still getting facial recognition systems before they receive guidelines on how to use them", notes a human rights activist who documents cases in Uttar Pradesh.

The Aadhaar Act of 2016 adds another layer of complexity. Initially intended to streamline welfare programs, it resulted in the creation of the world's largest biometric database. A rural schoolteacher, whose midday meal payments are linked to Aadhaar, explains, "Imagine giving every government clerk access to your fingerprints. Now, think about the lack of clear rules on who else can use that data". Recent reports indicate that banks and telecom companies are accessing Aadhaar information far beyond its original intent, a classic example of 'function creep' that the law fails to address.

For three years, India has been in discussions about the Personal Data Protection Bill, which promises safeguards similar to the GDPR. However, draft versions include exemptions that allow the government to bypass consent for "national security" reasons. A tech policy researcher in Bengaluru likens it to "building a house but leaving the master key with the builder". Meanwhile, various states are moving ahead: Delhi police are using facial recognition at protests, Telangana is tracking citizens through command centers, and Maharashtra is experimenting with AI-driven policing, all without specific laws to govern these technologies.

The human impact of these developments is evident in small-town courtrooms. Consider the case of an activist in Jharkhand who was wrongfully arrested due to a facial recognition error. His lawyer questions, "Where's the accountability when machines make mistakes?" Or look at the Muslim neighborhoods in Assam, where AI-driven "family tree" software disproportionately flags residents as suspected foreigners. A local imam expresses concern, saying, "The law should protect us from being reduced to data points, but it's not keeping pace".

India's legal framework currently resembles an old bicycle trying to navigate a superhighway. We have constitutional principles, like those established in the Puttaswamy case, but we lack detailed regulations. We possess advanced surveillance tools but insufficient oversight. As a retired police commissioner admits, "Even we don't fully understand the AI systems we're using." The answer isn't to abandon technology but to establish clear guidelines, laws on data minimization, mandatory impact assessments, and independent oversight bodies. Until that happens, every Indian is walking a digital tightrope, where their privacy hinges more on government discretion than on legal certainty. The real question isn't just about legality; it's about morality. Should a single mother selling vegetables at a train station be subjected to the same level of surveillance as a terror suspect? Can we trust algorithms to make that

distinction? These aren't just theoretical discussions, they're about the kind of society we're shaping for Ramesh, Priya, and millions of ordinary Indians navigating this new reality. In the narrow lanes of Old Delhi, a police constable squints at his smartphone screen. The grainy CCTV footage shows a crowded market, but the AI-powered facial recognition app has already circled three faces in red, potential matches for wanted criminals. "Earlier, we'd spend weeks chasing leads," he says, wiping sweat from his brow. "Now the computer gives us names in seconds". This scene repeats daily across India, where artificial intelligence is quietly revolutionizing law enforcement, for better and worse.

The National Automated Facial Recognition System (AFRS), launched in 2020, connects police stations across the country to a central database of "criminal faces". In theory, it helps find missing children and identify terrorists. In practice? A Lucknow-based lawyer shares the story of a client, a 22-year-old student arrested after the system falsely matched him to a robbery suspect. "It took us three months to prove his alibi", she says, flipping through court documents. "The police never questioned the technology's accuracy". Delhi's experience is particularly telling. During the 2020 riots, officers scanned over 50,000 faces from protest footage. Human rights groups found at least 32 wrongful arrests due to faulty matches, mostly poor migrants who couldn't afford lawyers. "The software struggles with Indian skin tones and crowded settings", explains an IIT professor who tested similar algorithms. "But no one audits these systems before deployment".

Beyond facial recognition, predictive policing is reshaping crime prevention. In Hyderabad, the Command Control Center uses AI to analyze years of crime data. Red dots glow on a digital map, "high-risk" areas where patrols are doubled. Sounds efficient? A local shopkeeper in one such "red zone" complains: "Since the software flagged our neighborhood, police stop and search my sons daily. Our crime? Being Muslim in the wrong postcode".

The Aarogya Setu app demonstrated how quickly emergency measures become permanent tools. Originally for COVID contact tracing, its location-tracking features were quietly repurposed by some states for general policing. "First they said it was for health, then for safety," shrugs a Mumbai taxi driver who still keeps the app installed, "now who knows?".

Perhaps most concerning are the shadowy partnerships between police and private tech firms. In Punjab, a U.S.-based company provided AI systems that allegedly labeled farmers' protest organizers as "potential extremists". No public tender, no transparency about how the algorithms work. "It's outsourcing bias," remarks a retired IPS officer. "Foreign companies train their AI on Western data, then sell it to Indian police who treat the results as gospel".

The human costs surface in unexpected ways. In a Chennai slum, women report avoiding women's health clinics after facial recognition cameras were installed outside. In Jaipur, street vendors dismantle stalls near newly installed "smart poles" with embedded cameras. "We're not criminals", says a flower seller, "but we feel like we're always being watched".

Yet there are glimpses of hope. The Telangana High Court recently ordered audits of police AI systems after a wrongful arrest. The Kerala government became the first to declare it won't use facial recognition at protests. Small victories, but they show the tide can turn.

As India stands at this technological crossroads, the fundamental question remains: Should crime-fighting tools decide who counts as a criminal? The answer will determine whether AI becomes a servant of justice or its unaccountable master. For now, the algorithms keep scanning, the databases keep growing, and ordinary Indians wonder, who's watching the watchers?

Global lessons, local realities: How India's AI surveillance stacks up against the world

From London's Ring of Steel to China's Social Credit System, every nation is grappling with the AI surveillance dilemma. But where does India fit in this global spectrum? This section examines how India's approach compares with both democratic benchmarks and authoritarian models revealing uncomfortable truths about our technological trajectory.

The Democratic Dilemma: India vs. Western Models

While the EU's GDPR mandates strict purpose limitation for surveillance tech, India's draft Data Protection Bill carves out sweeping national security exemptions. A telling contrast: When London police trialed live facial recognition, they published impact assessments and accepted 81% error rates. Meanwhile, Delhi Police deployed AFR systems without any public documentation on accuracy.

The U.S. Fourth Amendment requires warrants for most surveillance, whereas India's IT Act Section 69 allows bulk data collection under vague "public order" justifications. Even so, American cities like San Francisco banned police facial recognition after community protests—a civic pushback yet to gain momentum in India.

The China Parallel: Surveillance without safeguards

India's Safe City projects increasingly resemble China's Sharp Eyes program, with one crucial difference: China has laws (however problematic) governing AI surveillance use. Our systems operate in a legal vacuum, where even basic questions about data retention periods remain unanswered. However, India lacks China's centralized control—creating a dangerous patchwork. While Xinjiang's surveillance targets minorities systematically, in India, it's often local police using unvetted AI tools arbitrarily. A Hyderabad cyberlaw expert notes: "At least Chinese citizens know the rules. Here, the rules change depending on which police station you're dealing with".

Global South Comparisons: Lessons from Brazil & South Africa

Brazil's GDPR-inspired LGPD law shows developing nations can implement strong safeguards. Their facial recognition bans in São Paulo schools demonstrate community-centric tech governance.

South Africa's POPIA Act offers another model, requiring algorithmic impact assessments for public surveillance—a concept absent in India's policy debates. Notably, both countries involve civil society in tech oversight, unlike India's closed-door procurement processes.

The Swiss cheese Model: What India's missing Three critical gaps emerge:

- **Oversight Mechanisms:** Unlike the UK's surveillance camera commissioner or Canada's Privacy

Commissioner, India has no independent auditor for AI surveillance.

- **Accuracy Standards:** While Australia mandates 95%+ accuracy for police facial recognition, Indian vendors often supply untested algorithms.
- **Redress Systems:** The EU's right to explanation lets citizens challenge algorithmic decisions—a protection Indian courts haven't yet recognized.

As a Bengaluru tech ethicist puts it: "We've imported surveillance tools from Israel, AI models from America, and data practices from China, but forgot to import the safeguards each country developed through hard experience". This comparative analysis reveals India risks creating the worst of all worlds: Authoritarian scale surveillance with democratic-era accountability gaps. The path forward requires learning not just from global successes, but more importantly, from others' failures.

Conclusion: Charting India's path forward in the age of AI Surveillance

As India finds itself at a pivotal moment between technological progress and fundamental rights, the decisions made today will shape whether AI surveillance serves as a means of public safety or becomes a tool for social control. The evidence reveals a concerning paradox: while India's surveillance capabilities are on par with global superpowers, its legal protections are still rooted in a pre-digital age. From facial recognition systems mistakenly identifying protesters to predictive policing algorithms that reinforce caste biases, the human costs of unregulated AI are becoming increasingly evident.

A comparative analysis uncovers an uncomfortable reality: India is building a surveillance infrastructure akin to that of China, but with only superficial democratic safeguards. Unlike Western nations, where courts and legislatures actively limit surveillance overreach, or even authoritarian regimes with clear (if oppressive) rules, India's approach merges the worst aspects of both worlds, advanced technological capabilities paired with weak institutional oversight. This has led to what privacy expert's term "surveillance by inertia," where systems expand not through intentional policy but through bureaucratic momentum and vendor lobbying.

The path forward isn't about rejecting technology; it's about creating what Stanford researchers refer to as "constitutional AI", systems designed from the ground up to uphold fundamental rights. As discussions in chai shops from Srinagar to Thiruvananthapuram illustrate, ordinary Indians grasp this balance better than many policymakers: they desire safety, but not at the expense of dignity; security, but not through indiscriminate suspicion.

India's founding fathers enshrined privacy protections in the Constitution without foreseeing the advent of facial recognition algorithms. Today's challenge is to reinterpret those principles for the age of AI. The answer lies not in more technology, but in better governance, ensuring that as our cities become "smarter", our democracy becomes wiser in safeguarding what truly makes India humane: the freedom of its people to live without the fear of constant surveillance. The time for this reckoning is not tomorrow, but today, before the surveillance genie becomes impossible to contain.

Conclusion

India's embrace of AI-driven surveillance reflects both its ambition for security and the deep risks of unchecked technological power. The rapid deployment of facial recognition, predictive policing, and biometric systems without strong legal safeguards exposes citizens to errors, bias, and disproportionate targeting, threatening the very democratic values they are meant to protect. Unlike Western democracies with stronger oversight, or even authoritarian states with clear if restrictive rules, India risks combining high-tech surveillance with weak accountability, creating profound vulnerabilities. The way forward is not to abandon AI, but to ensure its regulation through robust privacy laws, independent oversight, and constitutional safeguards. Only then can India balance security with liberty in the digital era.

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